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Fredrik Berg Kjolstad

Assistant Professor, Computer Science

Academic Positions

March 2020 to present — **Assistant Professor**, *Stanford University*

Education

August 2011 to February 2020 — **Ph.D., Computer Science**, *Massachusetts Institute of Technology*

August 2009 to August 2011 — **M.S. Computer Science**, *University of Illinois at Urbana-Champaign*

August 2002 to June 2005 — **B.E., Computer Science**, *Norwegian University of Science and Technology in Gjøvik*

Publications

Conference and Journal Publications

PLDI 26 — A Compiler for Fused Relational Operations on Multisets. James Dong and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue PLDI*, 2026

PLDI 26 — Bonsai: Compiling Queries to Pruned Tree Traversals. Alexander J Root, Christophe Gyurgyik, Purvi Goel, Kayvon Fatahalian, Jonathan Ragan-Kelley, Andrew Adams, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue PLDI*, 2026

PLDI 26 — Decoupling Data Layouts from Bounding Volume Hierarchies. Christophe Gyurgyik, Alexander J Root, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue PLDI*, 2026

OSDI 26 — Optimal Software Pipelining and Warp Specialization for Tensor Core GPUs. Rupanshu Soi, Rohan Yadav, Fredrik Kjolstad, Alex Aiken, Maryam Dehnavi, Michael Garland, and Michael Bauer. *Proceedings of the Symposium on Operating System Design and Implementation*, 2026

OOPSLA 26 — Deegen: A JIT-Capable VM Generator for Dynamic Languages. Haoran Xu and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue OOPSLA*, 2026

ASPLOS 26 — FuseFlow: A Fusion-Centric Compilation Framework for Sparse Deep Learning on Streaming Dataflow. Rubens Lacouture, Nathan Zhang, Ritvik Sharma, Marco Siracusa, Fredrik Kjolstad, Kunle Olukotun, and Olivia Hsu. *International Conference on Architectural Support for Programming Languages and Operating Systems*, 2026

CGO 26 Fast Autoscheduling for Sparse ML Frameworks. Bobby Yan, Alexander J Root, Trevor Gale, David Broman, and Fredrik Kjolstad. *International Symposium on Code Generation and Optimization*, 2026

CGO 26 Ember: A Compiler for Embedding Operations on Decoupled Access-Execute Architectures. Marco Siracusa, Olivia Hsu, Victor Soria-Pardos, Joshua Randall, Arnaud Grasset, Eric Biscondi, Doug Joseph, Randy Allen, Fredrik Kjolstad, Miquel Moretó Planas, and Adrià Armejach. *International Symposium on Code Generation and Optimization*, 2026

OOPSLA 25 REPTILE: Performant Tiling of Recurrences. Muhammad Usman Tariq, Shiv Sundram, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 9, Issue OOPSLA*, 2025

MICRO 25 A Probabilistic Perspective on Tiling Sparse Tensor Algebra. Ritvik Sharma, Zi Xue, Nathan Zhang, Rubens Lacouture, Fredrik Kjolstad, Sara Achour, and Mark Horowitz. *Proceedings of the 58th IEEE/ACM International Symposium on Microarchitecture*, 2025

JSSC 25 Onyx: A 12-nm Programmable Accelerator for Dense and Sparse Applications. Kalhan Koul, Olivia Hsu, Yuchen Mei, Sai Gautham Ravipati, Maxwell Strange, Jackson Melchert, Alex Carsello, Taeyoung Kong, Po-Han Chen, Huifeng Ke, Keyi Zhang, Qiaoyi Liu, Gedeon Nyengele, Zhouhua Xie, Akhilesh Balasingam, Jayashree Adivarahan, Ritvik Sharma, Christopher Torng, Joel Emer, Fredrik Kjolstad, Mark Horowitz, and Priyanka Raina. *IEEE Journal of Solid-State Circuits*, 2025

IEEE Micro 25 Designing Programmable Accelerators for Sparse Tensor Algebra. Kalhan Koul, Zhouhua Xie, Maxwell Strange, Sai Gautham Ravipati, Bo Wun Cheng, Olivia Hsu, Po-Han Chen, Mark Horowitz, Fredrik Kjolstad, and Priyanka Raina. *IEEE Micro*, 2025

ASPLOS 25 Automatic Tracing in Task-Based Runtime Systems. Rohan Yadav, Michael Bauer, David Broman, Michael Garland, Alex Aiken, and Fredrik Kjolstad. *Architectural Support for Programming Languages and Operating Systems*, 2025

ASPLOS 25 Composing Distributed Computations Through Task and Kernel Fusion. Rohan Yadav, Shiv Sundram, Wonchan Lee, Michael Garland, Michael Bauer, Alex Aiken, and Fredrik Kjolstad. *Architectural Support for Programming Languages and Operating Systems*, 2025

CGO 25 Stardust: Compiling Sparse Tensor Algebra to a Reconfigurable Dataflow Architecture. Olivia Hsu, Alexander Rucker, Tian Zhao, Varun Desai, Kunle Olukotun, and Fredrik Kjolstad. *International Symposium on Code Generation and Optimization*, 2025

OOPSLA 24 Compiling Recurrences over Dense and Sparse Arrays. Shiv Sundram, Muhammad Usman Tariq, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 8, Issue OOPSLA*, 2024

OOPSLA 24 Compilation of Shape Operators on Sparse Arrays. Alexander J. Root, Bobby Yan, Peiming Liu, Christophe Gyurgyik, Aart J.C Bik, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 8, Issue OOPSLA*, 2024

OOPSLA 24 Compiler Support for Sparse Tensor Convolutions. Peiming Liu, Alexander J. Root, Anlun Xu, Yinying Li, Fredrik Kjolstad, and Aart J.C Bik. *Proceedings of the ACM on Programming Languages, Volume 8, Issue OOPSLA*, 2024

ISCA 24 The Dataflow Abstract Machine Simulator Framework. Nathan Zhang, Rubens Lacouture, Gina Sohn, Paul Mure, Qizheng Zhang, Fredrik Kjolstad, and Kunle Olukotun. *Proceedings of the International Symposium on Computer Architecture* (**Distinguished Artifact Award**), 2024

PLDI 24 Compilation of Modular and General Sparse Workspaces. Genghan Zhang, Olivia Hsu, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 8, Issue PLDI*, 2024

VLSI 24 Onyx: A 12nm 756 GOPS/W Coarse-Grained Reconfigurable Array for Accelerating Dense and Sparse Applications. Kalhan Koul, Maxwell Strange, Jackson Melchert, Alex Carsello, Yuchen Mei, Olivia Hsu, Taeyoung Kong, Po-Han Chen, Huifeng Ke, Keyi Zhang, Qiaoyi Liu, Gedeon Nyengele, Akhilesh Balasingam, Jayashree Adivarahan, Ritvik Sharma, Zhouhua Xie, Christopher Torng, Joel Emer, Fredrik Kjolstad, Mark Horowitz, Priyanka Raina. *IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, 2024

HPCA 24 Revet: A Language and Compiler for Dataflow Threads. Alexander Rucker, Shiv Sundram, Coleman Smith, Matt Vilim, Raghu Prabhakar, Fredrik Kjolstad, and Kunle Olukotun. *IEEE International Symposium on High-Performance Computer Architecture*, 2024

SC 23 Legate Sparse: Distributed Sparse Computing in Python. Rohan Yadav, Wonchan Lee, Melih Elibol, Manolis Papadakis, Taylor Lee-Patti, Michael Garland, Alex Aiken, Fredrik Kjolstad, and Michael Bauer. *ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis*, 2023

PLDI 23 Indexed Streams: A Formal Intermediate Representation for the Fused Execution of Contraction Operations. Scott Kovach, Praneeth Kolichala, Timothy Guo, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 7, Issue PLDI*, 2023

PLDI 23 Mosaic: An Interoperable Compiler for Tensor Algebra. Manya Bansal, Olivia Hsu, Kunle Olukotun, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 7, Issue PLDI* (**Distinguished Paper Award**), 2023

ASPLOS 23 BaCO: A Fast and Portable Bayesian Compiler Optimization Framework. Erik Hellsten, Artur Souza, Johannes Lenfers, Rubens Lacouture, Olivia Hsu, Adel Ejjeh, Fredrik Kjolstad, Michel Steuwer, Kunle Olukotun, and Luigi Nardi. *International Conference on Architectural Support for Programming Languages and Operating Systems*, 2023

ASPLOS 23 The Sparse Abstract Machine. Olivia Hsu, Maxwell Strange, Ritvik Sharma, Jaeyeon Won, Kunle Olukotun, Joel Emer, Mark Horowitz, and Fredrik Kjolstad. *Architectural Support for Programming Languages and Operating Systems*, 2023

TACO 23 Unified Buffer: Compiling Image Processing and Machine Learning Applications to Push-Memory Accelerators. Qiaoyi Liu, Jeff Setter, Dillon Huff, Maxwell Strange, Kathleen Feng, Mark Horowitz, Priyanka Raina, and Fredrik Kjolstad. *ACM Transactions on Architecture and Code Optimization (TACO)*, 2023

TECS 23 Aha: An agile approach to the design of coarse-grained reconfigurable accelerators and compilers. Kalhan Koul, Jackson Melchert, Kavya Sreedhar, Leonard Truong, Gedeon Nyengele, Keyi Zhang, Qiaoyi Liu, Jeff Setter, Po-Han Chen, Yuchen Mei, Maxwell Strange, Ross Daly, Caleb Donovick, Alex Carsello, Taeyoung Kong, Kathleen Feng, Dillon Huff, Ankita Nayak, Raj Setaluri, James Thomas, Nikhil Bhagdikar, David Durst, Zachary Myers, Nestan Tsiskaridze, Stephen Richardson, Rick Bahr, Kayvon Fatahalian, Pat Hanrahan, Clark Barrett, Mark Horowitz, Christopher Torng, Fredrik Kjolstad, and Priyanka Raina. *ACM Transactions on Embedded Computing Systems*, 2023

CGO 23 Looplets: A Language For Structured Coiteration. Willow Ahrens, Daniel Donenfeld, Fredrik Kjolstad, and Saman Amarasinghe. *International Symposium on Code Generation and Optimization*, 2023

SC 22 SpDISTAL: Compiling Distributed Sparse Tensor Computations. Rohan Yadav, Alex Aiken, and Fredrik Kjolstad. *ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis*, 2022

TACO 22 Compiler Support for Sparse Tensor Computations in MLIR. Aart J.C. Bik, Penporn Koanantakool, Tatiana Shpeisman, Nicolas Vasilache, Bixia Zheng, and Fredrik Kjolstad. *ACM Transactions on Architecture and Code Optimization*, 2022

PLDI 22 DISTAL: The Distributed Tensor Algebra Compiler. Rohan Yadav, Alex Aiken, and Fredrik Kjolstad. *ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2022

PLDI 22 Autoscheduling for Sparse Tensor Algebra with an Asymptotic Cost Model. Peter Ahrens, Fredrik Kjolstad, and Saman Amarasinghe. *ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2022

OOPSLA 21 Copy-and-Patch Compilation. Haoran Xu and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 5, Issue OOPSLA* (**Distinguished Paper Award**), 2021

OOPSLA 21 Compilation of Sparse Array Programming Models. Rawn Henry, Olivia Hsu, Rohan Yadav, Stephen Chou, Kunle Olukotun, Saman Amarasinghe, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 5, Issue OOPSLA*, 2021

OOPSLA 20 A Sparse Iteration Space Transformation Framework for Sparse Tensor Algebra. Ryan Senanayake, Changwan Hong, Ziheng Wang, Amalee Wilson, Stephen Chou, Shoaib Kamil, Saman Amarasinghe, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 4, Issue OOPSLA*, 2020

DAC 20 Rick Bahr, Clark Barrett, Nikhil Bhagdikar, Alex Carsello, Ross Daly, Caleb Donovick, David Durst, Kayvon Fatahalian, Kathleen Feng, Pat Hanrahan, Teguh Hofstee, Mark Horowitz, Dillon Huff, Fredrik Kjolstad, Taeyoung Kong, Qiaoyi Liu, Makai Mann, Jackson Melchert, Ankita Nayak, Aina Niemetz, Gedeon Nyengele, Priyanka Raina, Stephen Richardson, Raj Setaluri, Jeff Setter, Kavya Sreedhar, Maxwell Strange, James Thomas, Christopher Torng, Leonard Truong, Nestan Tsiskaridze, and Keyi Zhang. Creating an Agile Hardware Design Flow. *Design Automation Conference*, 2020

PLDI 20 Stephen Chou, Fredrik Kjolstad, and Saman Amarasinghe. Automatic Generation of Efficient Sparse Tensor Format Conversion Routines. *ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2020

CGO 19 Fredrik Kjolstad, Peter Ahrens, Shoaib Kamil, and Saman Amarasinghe. Sparse Tensor Algebra Optimization with Workspaces. *International Symposium on Code Generation and Optimization*, 2019

OOPSLA 18 Stephen Chou, Fredrik Kjolstad, and Saman Amarasinghe. Format Abstraction for Sparse Tensor Algebra Compilers. *Proceedings of the ACM on Programming Languages, Volume 2, Issue OOPSLA*, 2018

OOPSLA 17 Fredrik Kjolstad, Shoaib Kamil, Stephen Chou, David Lugato, and Saman Amarasinghe. The Tensor Algebra Compiler. *Proceedings of the ACM on Programming Languages, Volume 1, Issue OOPSLA* (**Distinguished Paper Award**), 2017

TOG 16 Fredrik Kjolstad, Shoaib Kamil Jonathan Ragan-Kelley, David I.W. Levin, Shinjiro Sueda, Desai Chen, Etienne Vouga, Danny M. Kaufman, Gurtej Kanwar, Wojciech Matusik, and Saman Amarasinghe. Simit: A Language for Physical Simulation. *ACM Transactions on Graphics (TOG, presented at SIGGRAPH)*, 2016

EuroMPI 13 Timo Schneider, Fredrik Kjolstad, and Torsten Hoefler. MPI Datatype Processing using Runtime Compilation. *The 20thEuropean MPI Users' Group Meeting (EuroMPI)* (**Best Paper Award**), 2013

ICSE 11 Fredrik Kjolstad, Danny Dig, Gabriel Acevedo, and Marc Snir. Transformation for Class Immutability. *33rd International Conference on Software Engineering*, 2011

Short Papers and Workshop Publications

PAW-ATM KDRSolvers: Scalable, Flexible, Task-Oriented Krylov Solvers. Proceedings of the Workshop on the Parallel Applications Workshop. David Zhang, Rohan Yadav, Alex Aiken, Fredrik Kjolstad, Sean Treichler. *Proceedings of the Workshop on the Parallel Applications Workshop, Alternatives to MPI+X*, 2025

SPAA Brief 20 Suzanne Mueller, Peter Ahrens, Stephen Chou, Fredrik Kjolstad, and Saman Amarasinghe. Sparse Tensor Transpositions. *ACM Symposium on Parallelism in Algorithms and Architectures (SPAA brief announcement)*, 2020

AVANCÉES 18 David Lugato, Fredrik Kjolstad, Stephen Chou, Saman Amarasinghe, and Shoaib Kamil. Taco: compilation et génération de code d'expressions tensorielles. *AVANCÉES No. 12*, 2018

ASE Tools 17 Fredrik Kjolstad, Stephen Chou, David Lugato, Shoaib Kamil, and Saman Amarasinghe. 32th IEEE/ACM International Conference on Automated Software Engineering. *taco: A Tool to Generate Tensor Algebra Kernels*, 2017

TOG Perspective 16 Gilbert Bernstein, and Fredrik Kjolstad. ACM Transactions on Graphics. *Why New Programming Languages for Simulation?*, 2016

PPoPP Short 12 Fredrik Kjolstad, Torsten Hoefler, and Marc Snir. Automatic Datatype Generation and Optimization. *17th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP short paper)*, 2012

CAP 10 Fredrik Kjolstad, Danny Dig, and Marc Snir. Bringing the HPC Programmer's IDE into the 21st Century through Refactoring. *SPLASH 2010 Workshop on Concurrency for the Application Programmer*, 2010

PPP 10 Fredrik Kjolstad and Marc Snir. Ghost Cell Pattern. *2nd Annual Workshop on Parallel Programming Patterns*, 2010

Awards

2025 Google ML and Systems Junior Faculty Award
2024 Stanford Tau Beta Pi Teaching Honor Roll
2024 Distinguished Artifact Award, ISCA
2023 Distinguished Paper Award, PLDI
2022 NSF CAREER Award
2021 Distinguished Paper Award, OOPSLA
2021 Google Research Scholar
2020 First Place MIT EECS George M. Sprowls PhD Thesis Award in Computer Science
2020 Robert N. Noyce Faculty Fellow, Stanford University School of Engineering
2017 Distinguished Paper Award, OOPSLA.
2016 Adobe Fellowship.
2013 Best Paper Award, EuroMPI

2011 Spring Grad Expo Best Poster Award from the University of Illinois at Urbana-Champaign

2006 Rosing Prize for best national IT-related student work in 2005 and 2006 from the Norwegian Computer Society

2005 Eureka Award for best bachelor project from the Norwegian University of Science and Technology in Gjøvik

Teaching

2020–present **CS343d Domain-Specific Programming Models and Compilers,** *Stanford University.*

2020–present **CS143 Compilers,** *Stanford University.*

Service

Program Committees

ASPLOS 2027 Program Committee
MLSys 2026 Program Committee
OOPSLA 2026 Review Committee
ASPLOS 2026 Program Committee
CACHP 2026 Program Committeee
Hot Chips 2025 Program Committee
PLDI 2025 Program Committee
POPL 2025 Program Committee
ASPLOS 2025 Program Committee
CGO 2025 Program Committee
ASPLOS 2024 Program Committee
OOPSLA 2024 Review Committee
GPCE 2024 Program Committee
CC 2024 Program Committee
PLDI 2023 Program Committee
PPOPP 2023 Program Committee
IPDPS 2023 Program Committee
CGO 2022 Program Committee
PPOPP 2021 Program Committee
PLDI 2021 Program Committee
SPLASH 2021 Student Research Competition Committee
PPOPP 2020 Program Committee
CGO 2020 Program Committee
PLDI 2020 Student Research Competition Committee
PLDI 2020 External Review Committee
ASPLOS 2020 External Review Committes

Chairs

PLDI 2025 Sponsorship co-chair
PLDI 2024 Sponsorship co-chair
PPoPP 2023 Publications chair

Journal and External Reviews

ACM TACO 2019, 2021, 2022, 2024
ACM TOMS 2021
ACM TOPC 2020
ACM TOG 2017
ACM SIGGRAPH 2019, 2020, 2024, 2026
ACM SIGGRAPH Asia 2023, 2024
IEEE Computer Architecture Letters 2022, 2023
IEEE TPDS 2017, 2021
NeurIPS 2023
Springer JPDC 2022

Organizer

2024 Sammaniversary
Workshop to celebrate the 60th birthday of Saman Amarasinghe. Hosted at MIT.

2024 Sparse Workshop (formerly CTSTA)
Workshop that brings together leading researchers on programming systems for sparse tensor algebra, graph algorithms, and relational algebra. Colocated with PLDI.

2023 Workshop on Compiler Techniques for Sparse Tensor Algebra
Workshop that brought together leading researchers on sparse tensor algebra compilation and computing. Colocated with PLDI.

2019 Invited Workshop on Compiler Techniques for Sparse Tensor Algebra
Invited workshop that brought together leading researchers on sparse tensor algebra compilation and computing from 11 universities, 6 companies and 3 national labs.

2012–2013 MIT Programming Languages Offsite Retreat Attended by seven CSAIL professors and their research groups. Re-organized program around many short talks, hosted panels, invited external speakers, and gave opening remarks.

Panels

NSF Review Panel (2021,2022)

PLDI PL Mentoring Workshop (June, 2020)

Invited Talks

January 2026 Portable Compilation for Future AI, *MLPerf-Bench Workshop.* Sydney, Australia

November 2025 Domain-Specific Software, Hardware, and their Composition, *UCSD Seminar.* La Jolla, CA

November 2025 Domain-Specific Software, Hardware, and their Composition, *Georgia Tech Guest Lecture.* Virtual

October 2025 Domain-Specific Software, Hardware, and their Composition, *LLNL Seminar.* Livermore, CA

September 2025 Portable Compilation in for Future AI, *Google Deepmind Seminar.* Mountain View, CA

September 2025 Portable Compilation for Sparse Computation, *Apple Seminar.* Cupertino, CA

August 2025 AI Kernel Programming Tutorial Introduction, *HotChips 2025.* Stanford, CA

August 2025 Portable Compilation in an Accelerated World, *Dagstuhl Workshop on Software Performance Engineering.* Schloss Dagstuhl, Germany

March 2025 Domain-Specific Software for ML, Hardware, and their Composition, *Conference on Parsimony and Learning* Stanford, CA

Feb. 2025 Domain-Specific Software, Hardware, and their Composition, *University of Illinois at Urbana-Champaign, Siebel School Colloquium Series.* Urbana, IL.

Jan. 2025 Four Languages for Portability, *Case Western Reserve University Lecture.* Virtual.

Oct. 2024 Four Languages for Portability, *UCLA PL Day.* Los Angeles, CA

Oct. 2024 Four Languages for Portability, *IBM.* Virtual.

Oct. 2024	Portable Compilation for Sparse Computation, *Qualcomm.* Virtual.
Sep. 2024	The Future of Sparse Computing is Compilers, *IEEE HPEC GraphBLAS BoF Keynote.* Virtual.
Sep. 2024	Four Languages for Portability, *KTH Digital Futures Hub.* Stockholm, Sweden.
Sep. 2024	Four Languages for Portability, *Forum on Specification and Design Languages (FDL) Keynote.* Stockholm, Sweden.
July. 2024	Hardware/Software Codesign for Sparse Neural Networks, *Dagstuhl Workshop on Resource-Efficient ML.* Schloss Dagstuhl, Germany.
Jun. 2024	Four Languages for Portability, *Sparse Workshop.* Copenhagen, Denmark.
Apr. 2024	Domain-Specific Software, Hardware, and their Composition, *UC Berkeley Slice Lab.* Berkeley, CA.
Oct. 2023	Hardware/Software Codesign For Sparse Neural Networks, *Meta AI & Systems Co-design Faculty Summit.* Menlo Park, CA.
Sep. 2023	Software and Hardware for Sparse ML. *CODAI Workshop,* Virtual Talk.
Jun. 2023	Portable Compilation for Sparse Computation. *PLDI DRAGSTERS Workshop Keynote,* Orlando, FL.
Jun. 2023	Software and Hardware for Sparse ML. *PLDI CTSTA Workshop,* Orlando, FL.
Apr. 2023	Portable Compilation for Sparse Computation. *Rice Future of Large-Scale Machine Learning Workshop Invited Talk,* Houston, TX.
Nov. 2022	Sparse Tensor Algebra Compilation Tutorial. *NVIDIA,* Virtual Event.
Oct. 2022	Portable Compilation for Sparse Computation. *UC Berkeley Sky Seminar,* Berkeley, CA.
Oct. 2022	Software and Hardware for Sparse ML. *Meta AI HW/SW Codesign Team,* Menlo Park, CA.
Oct. 2022	Portable Compilation of Sparse Computation. *LCPC Inited Talk,* Chicago, IL.
Aug. 2022	Software and Hardware for Sparse ML. *MLSys Chips & Compilers Symposium,* San Jose, CA.
Jun. 2022	Why you would want a compiler in your sparse library. *NVIDIA,* Virtual Event.
Apr. 2022	Sparse Tensor Algebra Compilation. *Meta PyTorch Team,* Menlo Park, CA.
Oct. 2021	Compiling Sparse Array Programming Languages. *Industry-Academia Partnership MIT/Harvard Cloud Workshop,* Virtual Event
Sep. 2021	Sparse Tensor Algebra Compilation. *Amazon Labs,* Cupertino, CA.
Jul. 2021	Fast Compilation and Sparse Compilation. *Google Brain Hardware,* Virtual Event.
Jun. 2021	Sparse Compilation and Fast Compilation. *Intel Research Overview,* Virtual Event.
Apr. 2021	Sparse Tensor Algebra Compilation. *Mathworks,* Virtual Event.
Mar. 2021	Sparse Tensor Algebra Compilation and Sparse Computation. *Accenture Labs Distinguished Researcher Talk,* Virtual Event.
Jan. 2020	Sparse Tensor Algebra Compilation. *SIAM Conference on Parallel Processing for Scientific Computing, invited talk,* Seattle, WA.
Jan. 2020	Sparse Tensor Algebra Optimization. *SIAM Conference on Parallel Processing for Scientific Computing, invited talk,* Seattle, WA.
Dec. 2019	Tensor Algebra Compilation with Workspaces. *Google Compilers and ML Reading Group,* Virtual Event.
Oct. 2019	Optimization in the Sparse Tensor Algebra Compiler. *LCPC Invited Talk,* Atlanta, GA.
Aug. 2019	The Sparse Tensor Algebra Compiler. *Norwegian University of Science and Technology AI Seminar,* Trondheim, Norway.
Jul. 2019	The Sparse Tensor Algebra Compiler. *MIT Fast Code Seminar*, Cambridge, MA
Jun. 2019	Abstraction without Friction. *NSF Workshop on Future Directions for Parallel and Distributed Computing*, Phoenix, AZ

May 2019 The Sparse Tensor Algebra Compiler. *MIT Graphics Group Seminar*, Cambridge, MA

Apr. 2019 The Sparse Tensor Algebra Compiler. *Cornell CS, Invited Seminar*, Ithaca, NY.

Apr. 2019 The Sparse Tensor Algebra Compiler. *Stanford CS, Invited Seminar*, Stanford, CA.

Mar. 2019 The Sparse Tensor Algebra Compiler. *Georgia Tech CSE, Invited Seminar,* Atlanta, GA.

Feb. 2019 The Sparse Tensor Algebra Compiler. *Innovations in Software Engineering Conference, Invited Talk,* Pune, India

Jan. 2019 The Sparse Tensor Algebra Compiler. *UT Austin ECE,* Austin, TX.

Sep. 2018 The Tensor Algebra Compiler. *University of Texas,* Austin, TX.

Jun. 2018 The Sparse Tensor Algebra Compiler. *Adobe Research,* Seattle, WA.

Jun. 2018 The Sparse Tensor Algebra Compiler. *NVIDIA,* Redmond, WA.

Jun. 2018 The Sparse Tensor Algebra Compiler. *University of Washington,* Seattle, WA.

Jun. 2018 The Sparse Tensor Algebra Compiler. *Facebook AI,* Menlo Park, CA.

May 2018 The Sparse Tensor Algebra Compiler. *Stanford,* Stanford, CA.

May 2018 The Sparse Tensor Algebra Compiler. *UC Berkeley,* Stanford, CA.

May 2018 The Sparse Tensor Algebra Compiler. *Google Brain,* Mountain View, CA.

Apr. 2018 The Tensor Algebra Compiler. *Industry-Academia Partnership MIT Cloud Workshop,* Cambridge, MA.

Mar. 2018 The Tensor Algebra Compiler. *SIAM Conference on Parallel Processing for Scientific Computing,* Tokyo, Japan

Nov. 2017 The Tensor Algebra Compiler and Simit. *University of Illinois,* Urbana-Champaign, IL.

Oct. 2017 The Tensor Algebra Compiler. *Microsoft Research,* Redmond, WA.

Sep. 2016 Simit: A Language for Computing on Sparse Systems. *Intel Research*, Hudson, MA.

Aug. 2016 Simit: A Language for Computing on Sparse Systems. *Microsoft Research*, Redmond, WA.

Oct. 2015 Simit: A Language for Computing on Sparse Systems. *MIT Computer Graphics Group Annual Retreat,* Beverly, MA.

Oct. 2008 Performance Optimization of Embedded 3D Graphics Applications. *ARM Developer's Conference,* Santa Clara, CA.

PhD Students

James Dong

Christophe Gyurgyik

Scott Kovach

Rubens Lacouture (co-advised with Kunle Olukotun)

Alexander Root

Shiv Sundram

Haoran Xu

Rohan Yadav (co-advised with Alex Aiken)

Bobby Yan

Alumni

Olivia Hsu (co-advised with Kunle Olukotun), faculty at CMU.

Trevor Gale (co-advised with Matei Zaharia), research Scientist at Google Deepmind.

Work Experience

Feb 2026 to current	**Gimlet Labs**, *Member of Technical Staff*
October 2024 to December 2025	**Google Deepmind**, *Visiting Researcher*
November 2022 to February 2024	**Amazon**, *Amazon Visiting Academic*
January 2007 to August 2009	**ARM Ltd.**, *Graphics Software Engineer*
August 2006 to December 2006	**Accenture Technology Solutions**, *Programmer*